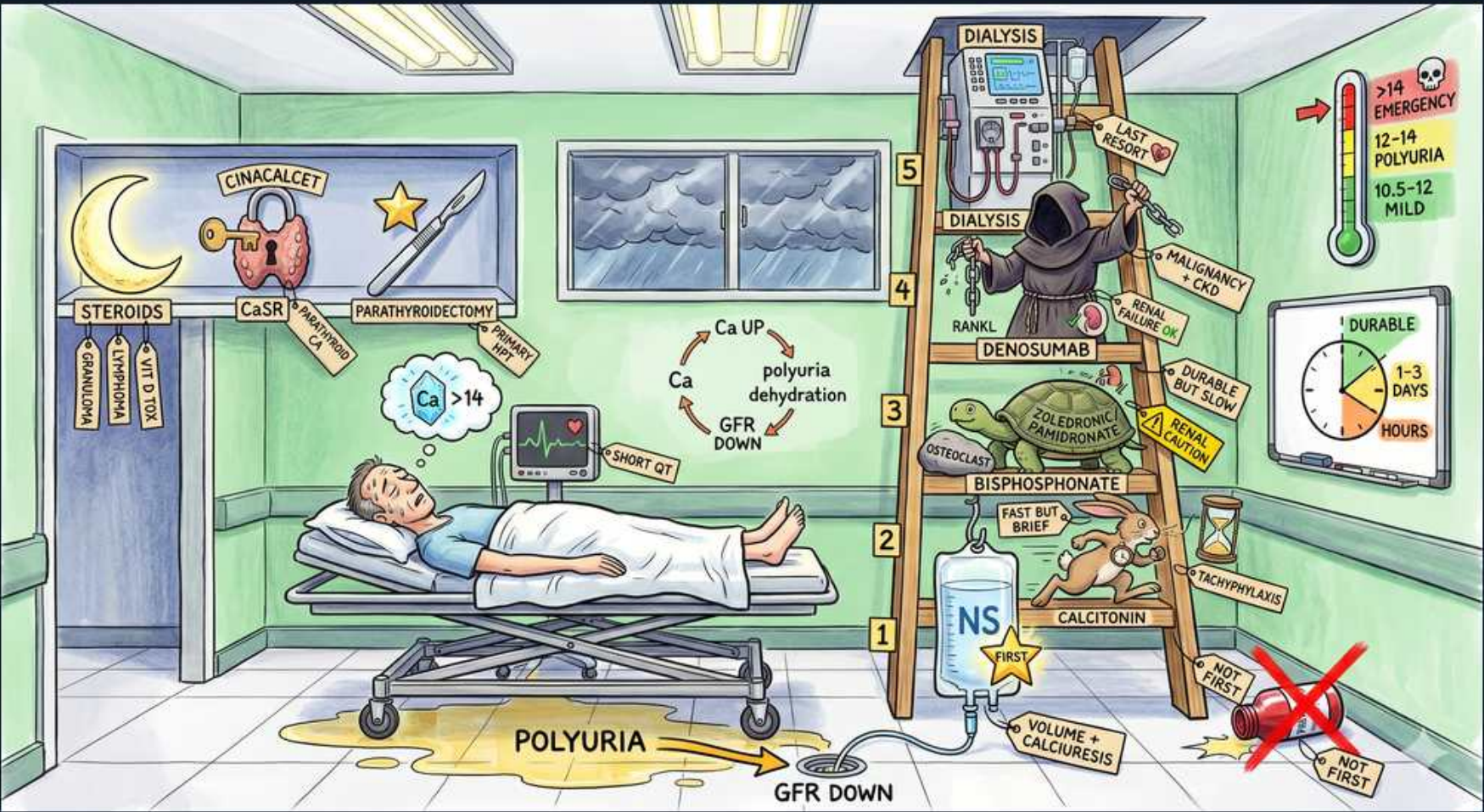


Calcium/PTH/VitD — The Calcium Crisis ER Ladder



1. Steroids (granuloma/lymphoma)

WHAT YOU SEE

A 'STEROIDS' box on the wall left.

WHAT IT ENCODES

Glucocorticoids (e.g., prednisone 20-40 mg/day or hydrocortisone 100 mg IV q6h) lower calcium specifically in calcitriol-mediated hypercalcemia. In granulomatous disease (sarcoidosis, TB, fungal) and some lymphomas, activated macrophages express 1-alpha-hydroxylase, converting 25-OH vitamin D to active 1,25-(OH)₂ vitamin D independent of PTH, driving gut calcium absorption. Steroids suppress this extrarenal 1-alpha-hydroxylase and reduce intestinal absorption; effect takes 2-5 days.

BOARD TRAP

A board case gives an SCC of the lung with hypercalcemia and a low PTH, then offers 'glucocorticoids' as the answer — WRONG. Steroids only work for calcitriol-driven (vitamin D-mediated) hypercalcemia such as sarcoid or lymphoma; they do NOT lower calcium in PHPT or PTHrP-mediated (humoral) malignancy. The discriminator is the mediator: check 1,25-OH vitamin D (high → steroids work) versus PTHrP (high → use saline/bisphosphonate).

2. Cinacalcet (CaSR)

WHAT YOU SEE

A padlock labeled 'CINACALCET / CaSR'.

WHAT IT ENCODES

Cinacalcet is a calcimimetic that allosterically activates the calcium-sensing receptor (CaSR) on parathyroid cells, increasing the gland's sensitivity to calcium so it secretes LESS PTH, thereby lowering serum calcium. It is used for hypercalcemia of parathyroid carcinoma, inoperable/refractory PHPT, and secondary hyperparathyroidism in dialysis patients. It is an oral chronic agent, not an acute crisis drug; a key side effect is hypocalcemia.

BOARD TRAP

A patient in acute hypercalcemic crisis (Ca 15, confused, dehydrated) is offered 'start cinacalcet' — WRONG. Cinacalcet is oral and slow, not for the emergency; the crisis answer is IV normal saline first. Cinacalcet's niche is chronic management of parathyroid carcinoma or non-operable PHPT, not the ER ladder.

3. Parathyroidectomy

WHAT YOU SEE

A scalpel/star labeled 'PARATHYROIDECTOMY'.

WHAT IT ENCODES

Parathyroidectomy is the only definitive cure for PHPT-driven hypercalcemia, removing the adenoma (85%), hyperplastic glands (4-gland, often MEN1/MEN2A), or carcinoma. Intraoperative PTH should fall >50% from baseline within 10 minutes of excision (Miami criterion), confirming the offending gland is out. It addresses the cause; the rest of the ladder only treats the calcium acutely.

BOARD TRAP

A stable outpatient with PHPT meeting surgical criteria is offered 'IV bisphosphonate' as definitive management — WRONG. Bisphosphonates and cinacalcet are temporizing/medical options; surgery is the only cure. The discriminator: if criteria are met and the patient is operable, parathyroidectomy is the definitive answer.

4. Ca >14 / short QT

WHAT YOU SEE

A monitor showing Ca >14 and 'SHORT QT'.

WHAT IT ENCODES

Severe hypercalcemia (Ca >14 mg/dL or any symptomatic level) is a medical emergency. Calcium accelerates cardiac myocyte repolarization, SHORTENING the QT interval (specifically the ST segment / QTc); extreme levels can cause Osborn-like changes and arrhythmia. Symptoms follow 'stones, bones, abdominal groans, psychiatric overtones' plus polyuria and dehydration.

BOARD TRAP

An ECG showing a SHORT QT in a hypercalcemic patient is misread as 'consistent with hypocalcemia' — WRONG and reversed. Hypercalcemia SHORTENS QT; hypocalcemia PROLONGS QT (risk of torsades). The discriminator is direction: high Ca = short QT, low Ca = long QT.

5. Severity thermometer

WHAT YOU SEE

A thermometer: '>14 EMERGENCY, 12-14 POLYURIA, 10.5-12 MILD'.

WHAT IT ENCODES

Calcium severity tiers guide urgency: mild 10.5-12 mg/dL (often asymptomatic, treat cause), moderate 12-14 mg/dL (polyuria, dehydration, may treat per symptoms), severe/crisis >14 mg/dL (always aggressive treatment regardless of symptoms). Always interpret against albumin (corrected Ca = measured + 0.8 x [4 - albumin]) or measure ionized calcium.

BOARD TRAP

A reported total Ca of 11.5 with albumin 2.0 is called 'only mild hypercalcemia, observe' — WRONG; corrected Ca = 11.5 + 0.8 x (4-2) = 13.1, which is moderate. Conversely, a low-albumin patient with 'normal' total Ca may actually be hypercalcemic. Always correct for albumin before tiering.

6. Dialysis (top rung)

WHAT YOU SEE

A dialysis machine at the top of the ladder labeled 'DIALYSIS'.

WHAT IT ENCODES

Hemodialysis with a low- or zero-calcium dialysate is the rescue/last-line therapy for life-threatening hypercalcemia, removing calcium directly when other measures fail or cannot be used. It is the answer when the patient has renal failure or heart failure (can't tolerate aggressive saline), or has refractory crisis-level Ca with neurologic compromise. Onset is immediate during the session.

BOARD TRAP

A hypercalcemic crisis patient with anuric ESRD and pulmonary edema is offered 'aggressive 4-6 L normal saline' — WRONG/dangerous, since they can't excrete the volume and will flood. The discriminator: renal failure or volume-overload + crisis-level Ca = hemodialysis, not saline.

7. Denosumab

WHAT YOU SEE

A 'DENOSUMAB' rung with RANKL.

WHAT IT ENCODES

Denosumab is a monoclonal antibody against RANKL that blocks osteoclast formation/activity, lowering bone resorption and serum calcium. Dose for hypercalcemia of malignancy is 120 mg SC. Its key advantage: NO renal dose adjustment, making it the agent of choice when GFR is too low for bisphosphonates or in bisphosphonate-refractory hypercalcemia. Watch for hypocalcemia, especially with low vitamin D.

BOARD TRAP

A malignancy-hypercalcemia patient with eGFR 25 is offered 'IV zoledronic acid' — WRONG, since bisphosphonates are nephrotoxic and contraindicated/risky at low GFR. The discriminator: renal impairment or bisphosphonate failure → denosumab (no renal adjustment) is the correct calcium-lowering antiresorptive.

8. Bisphosphonate (turtle = durable but slow)

WHAT YOU SEE

A turtle labeled 'BISPHOSPHONATE — durable but slow, 2-4 days'.

WHAT IT ENCODES

IV bisphosphonates inhibit osteoclast-mediated bone resorption and are the MAINSTAY for sustained lowering in hypercalcemia of malignancy. Zoledronic acid 4 mg IV over 15 min is preferred (more potent than pamidronate 60-90 mg). Onset is slow (2-4 days) but durable (effect lasts 2-4 weeks). Adjust/avoid in renal impairment; risks include nephrotoxicity, hypocalcemia, and osteonecrosis of the jaw.

BOARD TRAP

A crashing patient with Ca 16 and confusion is given 'IV zoledronic acid' as the FIRST and only move — WRONG. Bisphosphonates take 2-4 days to peak, so they do nothing for the acute crisis. The discriminator: give saline first (and calcitonin to bridge), then add a bisphosphonate for durable control; the turtle is slow but lasting.

9. Calcitonin (rabbit = fast but brief)

WHAT YOU SEE

A rabbit labeled 'CALCITONIN — fast but brief'.

WHAT IT ENCODES

Calcitonin (4-8 IU/kg SC/IM q6-12h) inhibits osteoclasts AND increases renal calcium excretion, dropping calcium by 1-2 mg/dL within 4-6 hours — the fastest agent on the ladder. It serves as a BRIDGE while the bisphosphonate takes effect over 2-4 days. Tachyphylaxis (downregulation of receptors) limits useful effect to ~48 hours.

BOARD TRAP

Calcitonin is chosen as MONOTHERAPY for durable control of malignant hypercalcemia — WRONG. Its effect is fast but brief (tachyphylaxis at ~48h); it must be paired with saline plus a bisphosphonate/denosumab for lasting control. The discriminator: rabbit (fast, short) bridges to the turtle (slow, durable).

10. Normal saline (bottom rung, FIRST)

WHAT YOU SEE

An NS IV bag at rung 1, labeled 'NS'.

WHAT IT ENCODES

Aggressive isotonic (0.9%) saline, typically 200-300 mL/hr to maintain urine output ~100-150 mL/hr (often 4-6 L/day), is the FIRST-line treatment for hypercalcemia. It corrects the near-universal dehydration, restores GFR, and increases urinary calcium excretion (volume expansion reduces proximal tubular Na⁺ and passive Ca⁺⁺ reabsorption). Monitor for volume overload, especially in heart/renal failure.

BOARD TRAP

A textbook sequencing trap offers 'furosemide' or 'IV bisphosphonate' as the first step in acute hypercalcemia — WRONG. The very first move is always IV normal saline to rehydrate; only after volume repletion are other agents (calcitonin to bridge, bisphosphonate for durability) added. The discriminator: fluids first, every time.

11. Volume → calciuresis

WHAT YOU SEE

A star labeled 'VOLUME → CALCIURESIS'.

WHAT IT ENCODES

Restoring intravascular volume is the core mechanism of saline therapy: expansion increases GFR and reduces proximal tubular sodium (and thus passively coupled calcium) reabsorption, markedly increasing urinary calcium excretion (calciuresis). This both dilutes serum calcium and actively dumps it in the urine, breaking the dehydration-hypercalcemia vicious cycle.

BOARD TRAP

A learner reasons that since 'loop diuretics increase calcium excretion,' they should be added early to boost calciuresis — WRONG. The dominant driver of calciuresis is VOLUME repletion, not the diuretic; giving a loop before rehydration worsens the dehydration that caused the hypercalcemia. Restore volume first.

12. Furosemide NOT first (red X)

WHAT YOU SEE

A red X over a loop diuretic, 'NOT FIRST'.

WHAT IT ENCODES

Loop diuretics (furosemide) block the Na-K-2Cl transporter, reducing the lumen-positive gradient that drives calcium reabsorption in the thick ascending limb, thereby promoting calciuresis. However, they are NO LONGER routine in hypercalcemia and are reserved for patients who become volume-overloaded after saline (e.g., heart/renal failure). Using them early causes volume depletion that paradoxically worsens hypercalcemia.

BOARD TRAP

The classic outdated-teaching trap: 'give furosemide to treat hypercalcemia' as an early/first step — WRONG. Early loop diuretics deplete volume and worsen hypercalcemia. They are only appropriate AFTER the patient is volume-replete and shows signs of overload. The discriminator: saline first; furosemide only for fluid overload, never as the opener.

13. Hypercalcemia symptom triad ($\text{Ca} \uparrow \rightarrow \text{polyuria} \rightarrow \text{GFR} \downarrow$)

WHAT YOU SEE

Center text: Ca UP $\rightarrow$ polyuria/dehydration $\rightarrow$ GFR DOWN.

WHAT IT ENCODES

Hypercalcemia activates the renal CaSR and antagonizes ADH action in the collecting duct, causing nephrogenic diabetes insipidus $\rightarrow$ polyuria $\rightarrow$ dehydration $\rightarrow$ falling GFR. The falling GFR reduces calcium clearance, raising calcium further: a self-perpetuating vicious cycle. This is exactly why IV saline (rehydration) is the priority that breaks the loop.

BOARD TRAP

A polyuric, hypercalcemic patient is worked up for primary diabetes insipidus or started on desmopressin — WRONG framing. The polyuria here is nephrogenic, driven by the hypercalcemia itself; treating the calcium (with saline) resolves it. The discriminator: correct the calcium, not chase the urine output with ADH.

14. Polyuria / GFR down (floor)

WHAT YOU SEE

A yellow puddle: 'POLYURIA' and arrow 'GFR DOWN'.

WHAT IT ENCODES

The yellow puddle depicts the consequence: hypercalcemia-induced polyuria leads to dehydration and falling GFR, which concentrates serum calcium and impairs its renal clearance. This is the pathophysiologic justification for fluids being rung #1 — you must refill the tank before any other agent can work efficiently.

BOARD TRAP

A dehydrated, prerenal-AKI hypercalcemic patient is held off fluids 'because the creatinine is rising' — WRONG. The rising creatinine is from hypovolemia caused by the hypercalcemia, and saline both restores GFR and lowers calcium. The discriminator: in hypercalcemic AKI, the treatment for the kidney IS the saline, not fluid restriction.